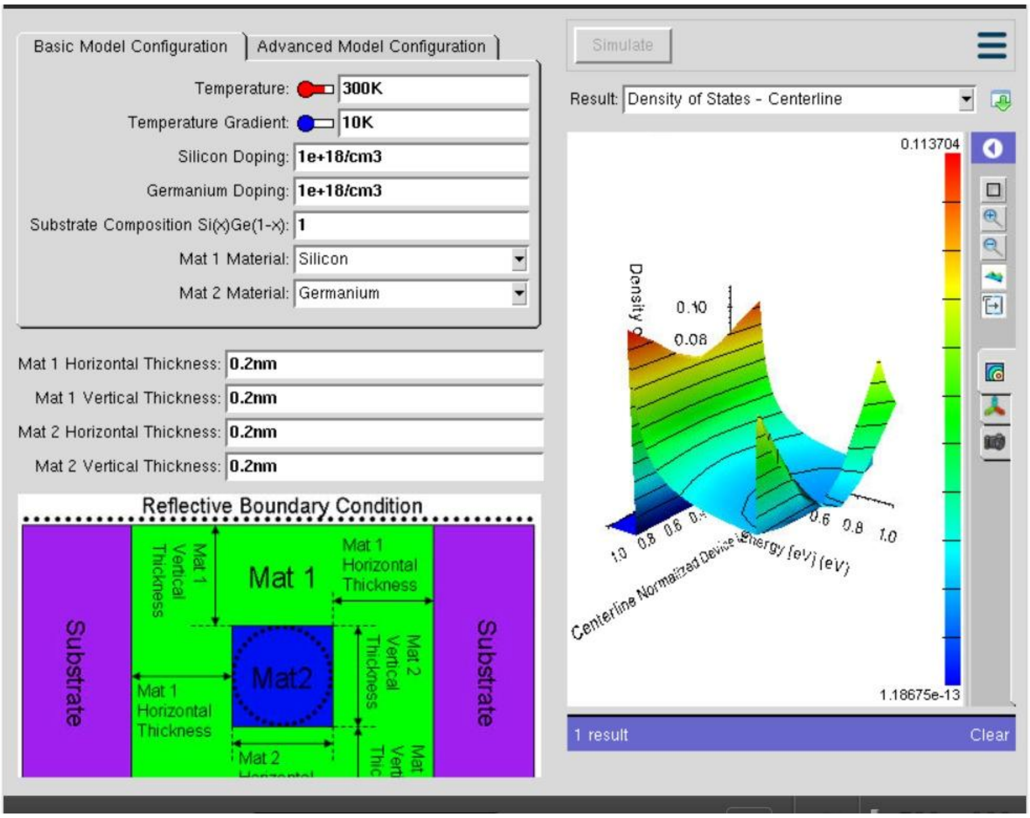
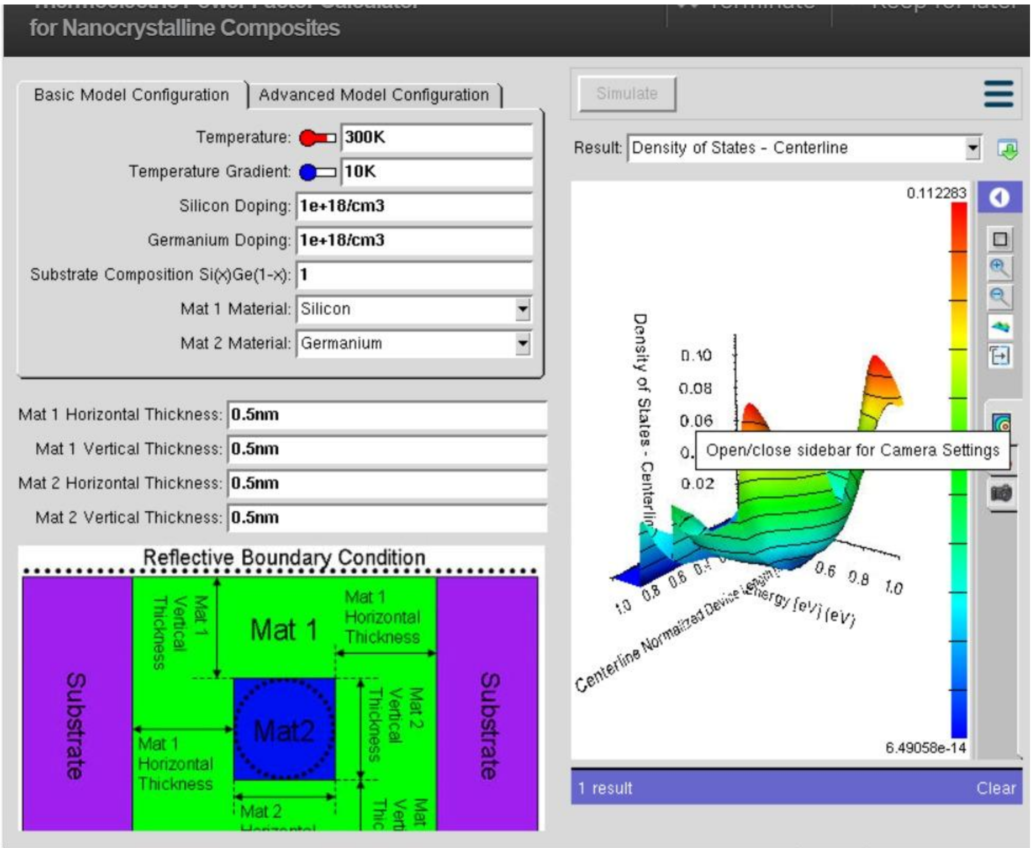
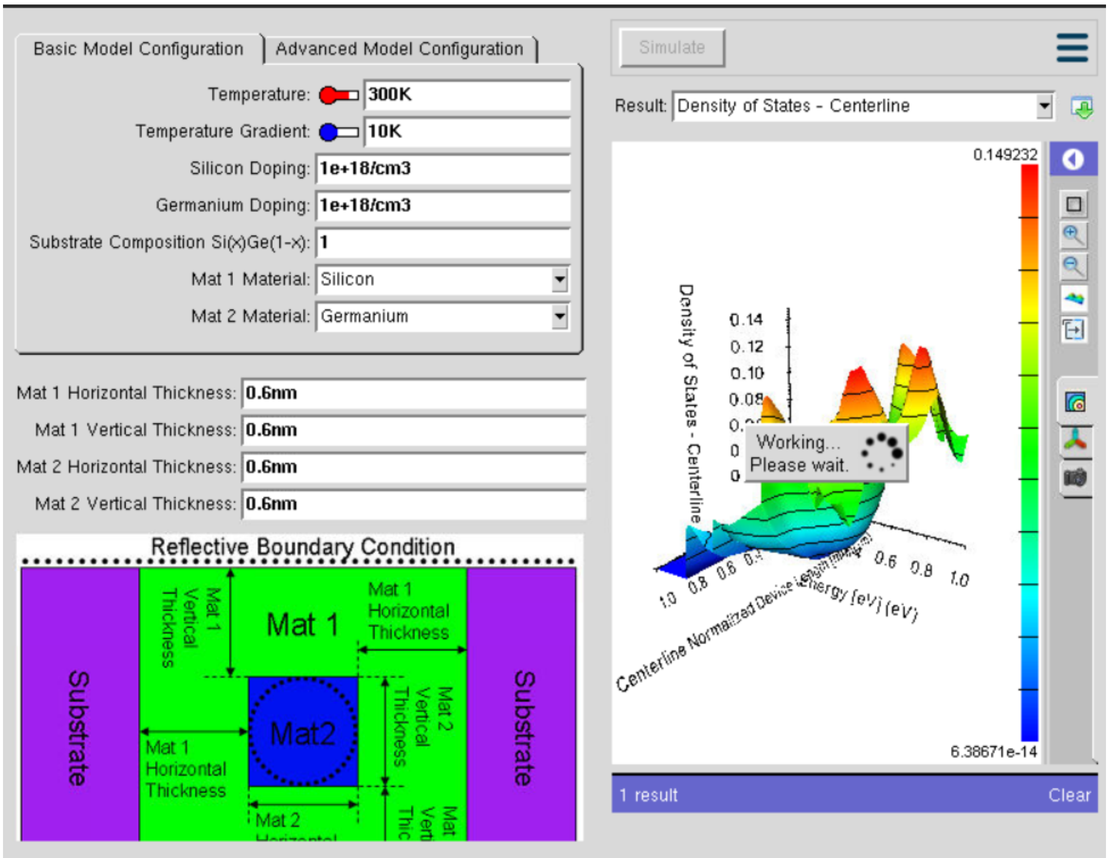
2nd Experiment:



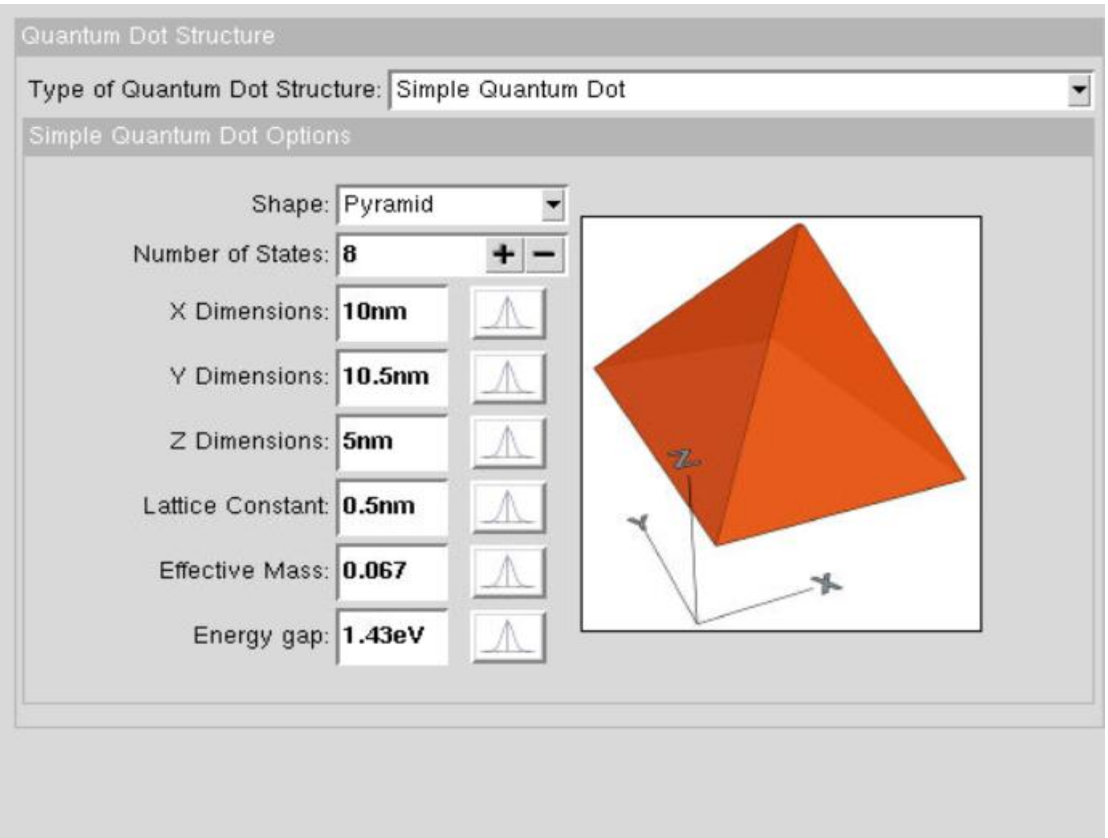
Case Study 1:

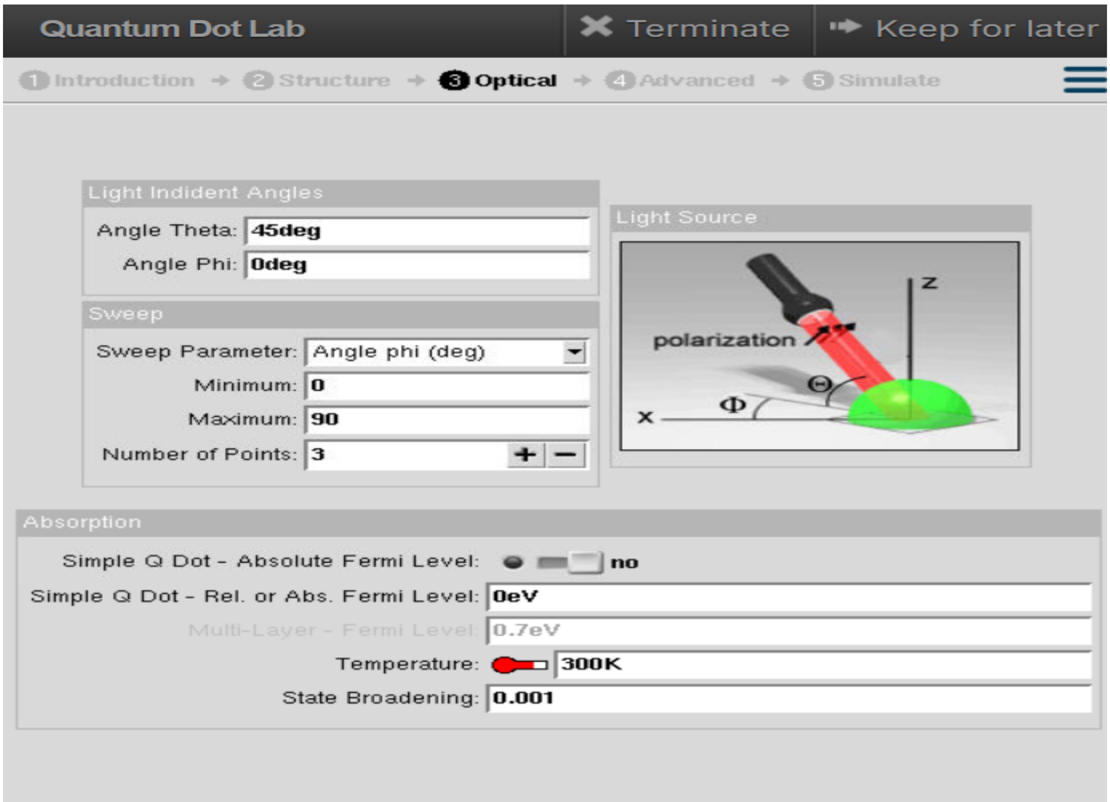


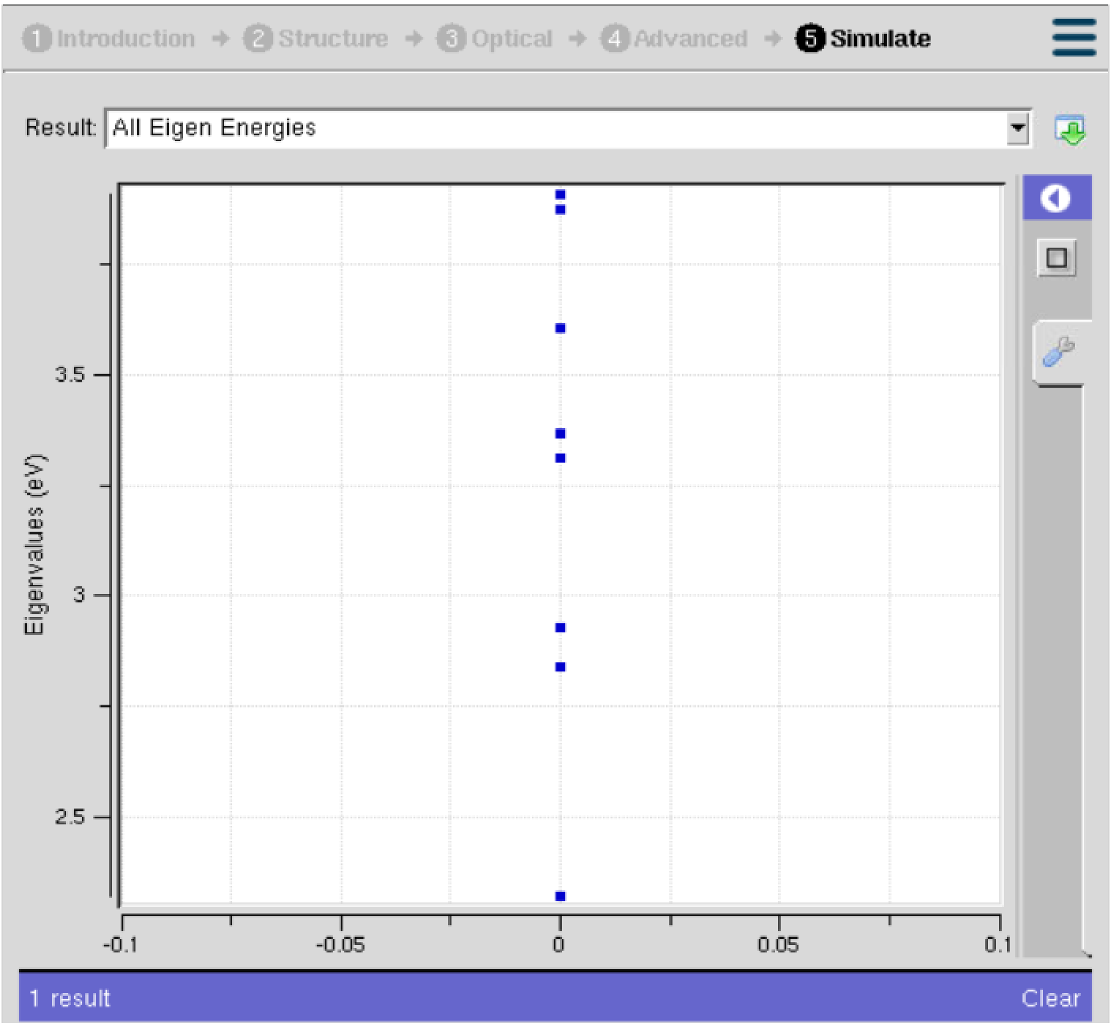
Case Study 2:



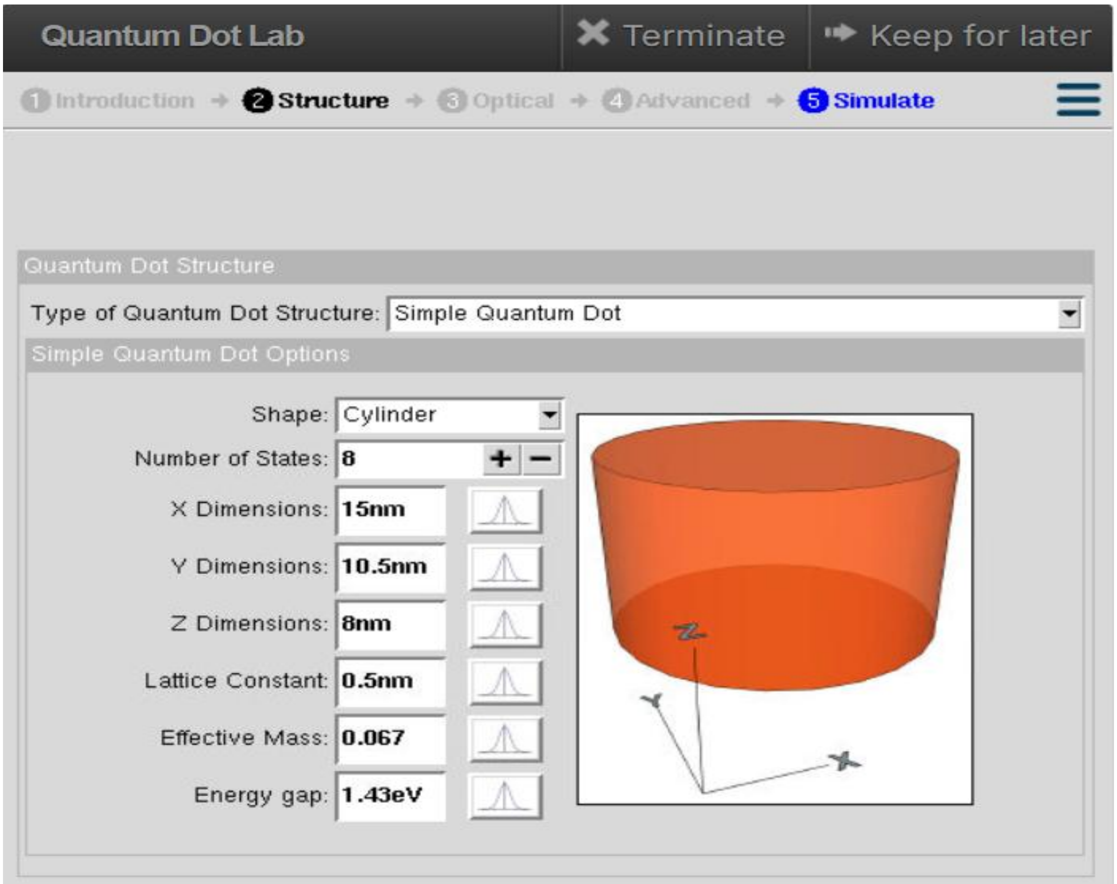
3rd Experiment

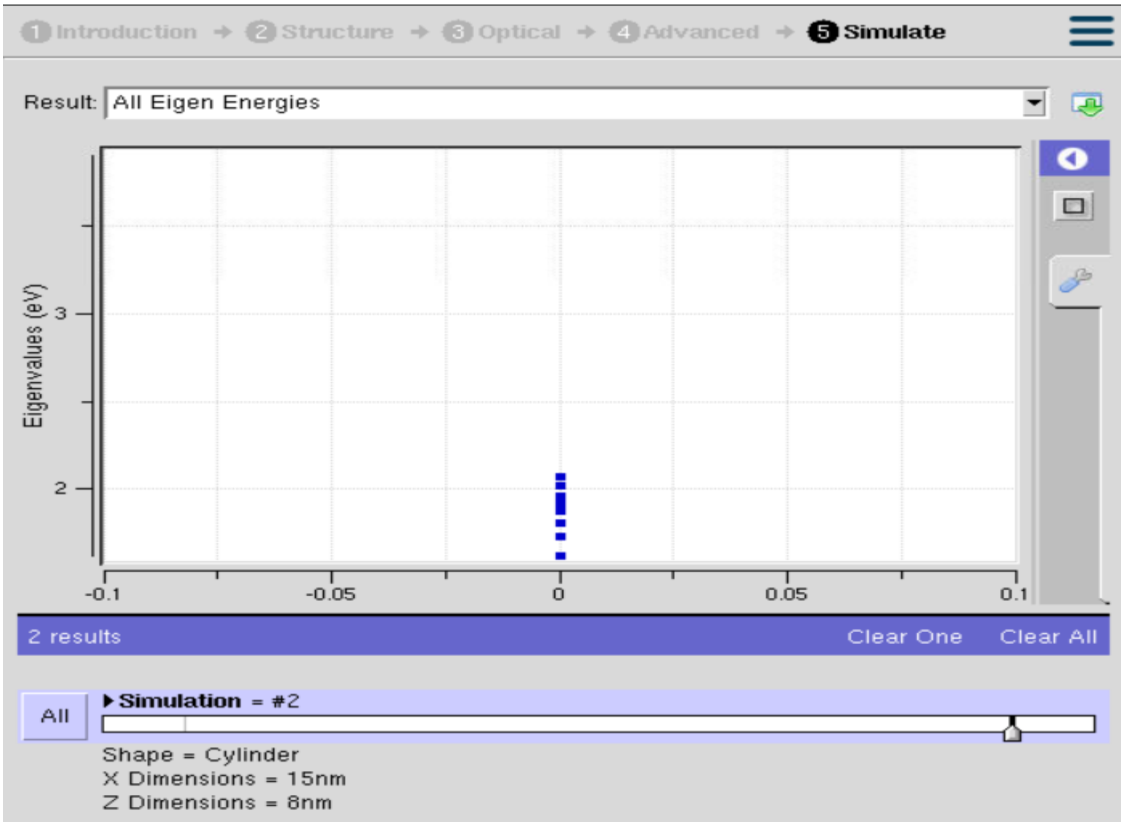



Quantum Dot Lab
Terminate
Keep for later
1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Light Incident Angles
Angle Theta: 45deg
Angle Phi: 0deg
Sweep
Sweep Parameter: Angle phi (deg)
Minimum: 0
Maximum: 90
Number of Points: 3
Light Source
z
polarization
Θ
Φ
x
Absorption
Simple Q Dot - Absolute Fermi Level: no
Simple Q Dot - Rel. or Abs. Fermi Level: 0eV
Multi-Layer - Fermi Level: 0.7eV
Temperature: 300K
State Broadening: 0.001

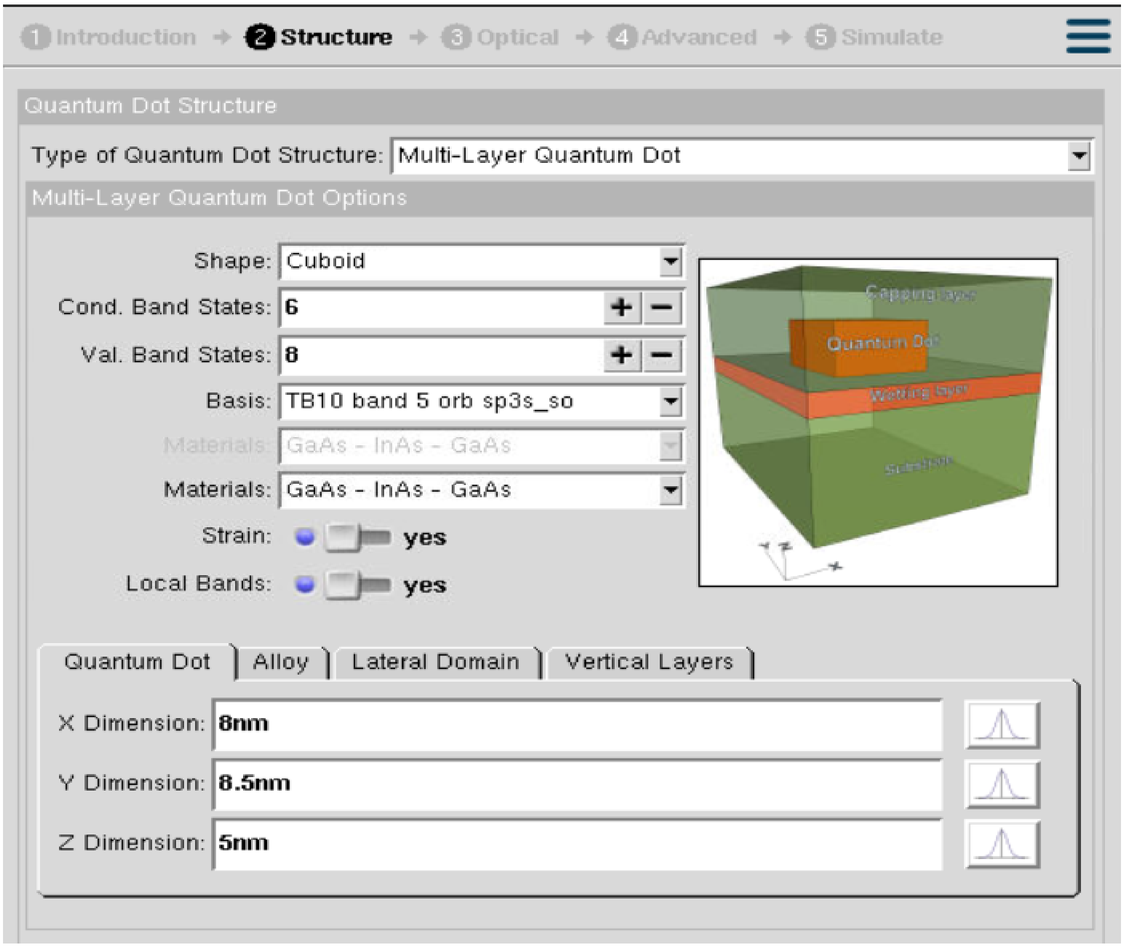

1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Result: All Eigen Energies
Eigenvalues (eV)
3.5
3
2.5
-0.1
-0.05
0
0.05
0.1
1 result
Clear

Case Study 1

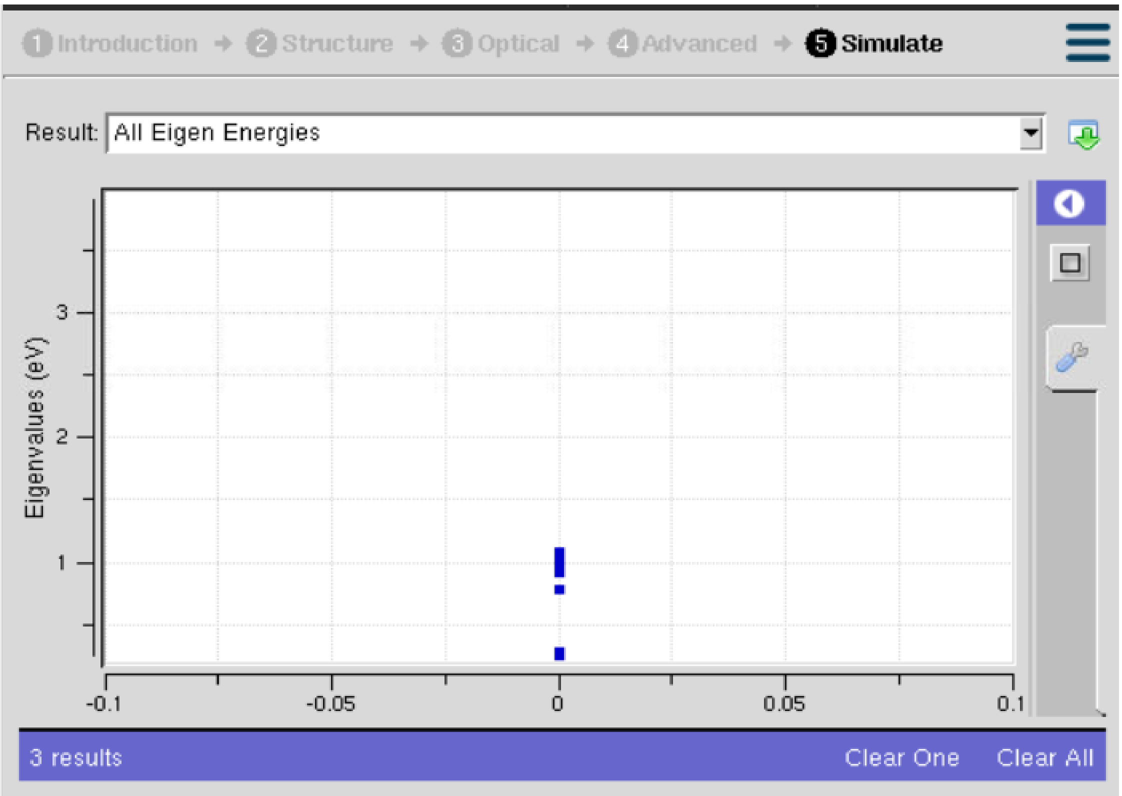

Quantum Dot Lab
Terminate
Keep for later
1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure: Simple Quantum Dot
Simple Quantum Dot Options
Shape: Cylinder
Number of States: 8
X Dimensions: 15nm
Y Dimensions: 10.5nm
Z Dimensions: 8nm
Lattice Constant: 0.5nm
Effective Mass: 0.067
Energy gap: 1.43eV


1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Result: All Eigen Energies
Eigenvalues (eV)
3
2
-0.1
-0.05
0
0.05
0.1
2 results
Clear One
Clear All
All
Simulation = #2
Shape = Cylinder
X Dimensions = 15nm
Z Dimensions = 8nm

Case Study 2

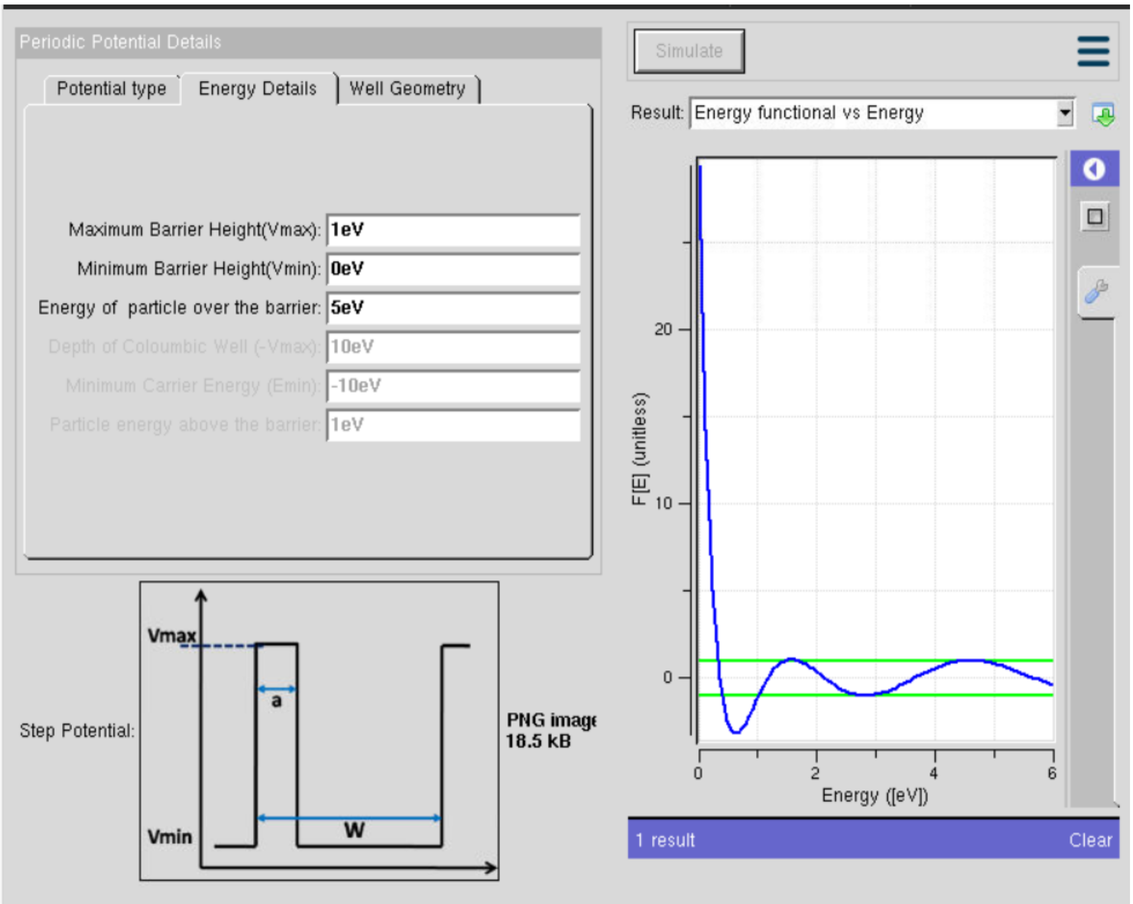


1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure: Multi-Layer Quantum Dot
Multi-Layer Quantum Dot Options
Shape: Cuboid
Cond. Band States: 6
Val. Band States: 8
Basis: TB10 band 5 orb sp3s_so
Materials: GaAs - InAs - GaAs
Materials: GaAs - InAs - GaAs
Strain: yes
Local Bands: yes
Capping layer
Quantum Dot
Wetting layer
Substrate
Quantum Dot
Alloy
Lateral Domain
Vertical Layers
X Dimension: 8nm
Y Dimension: 8.5nm
Z Dimension: 5nm

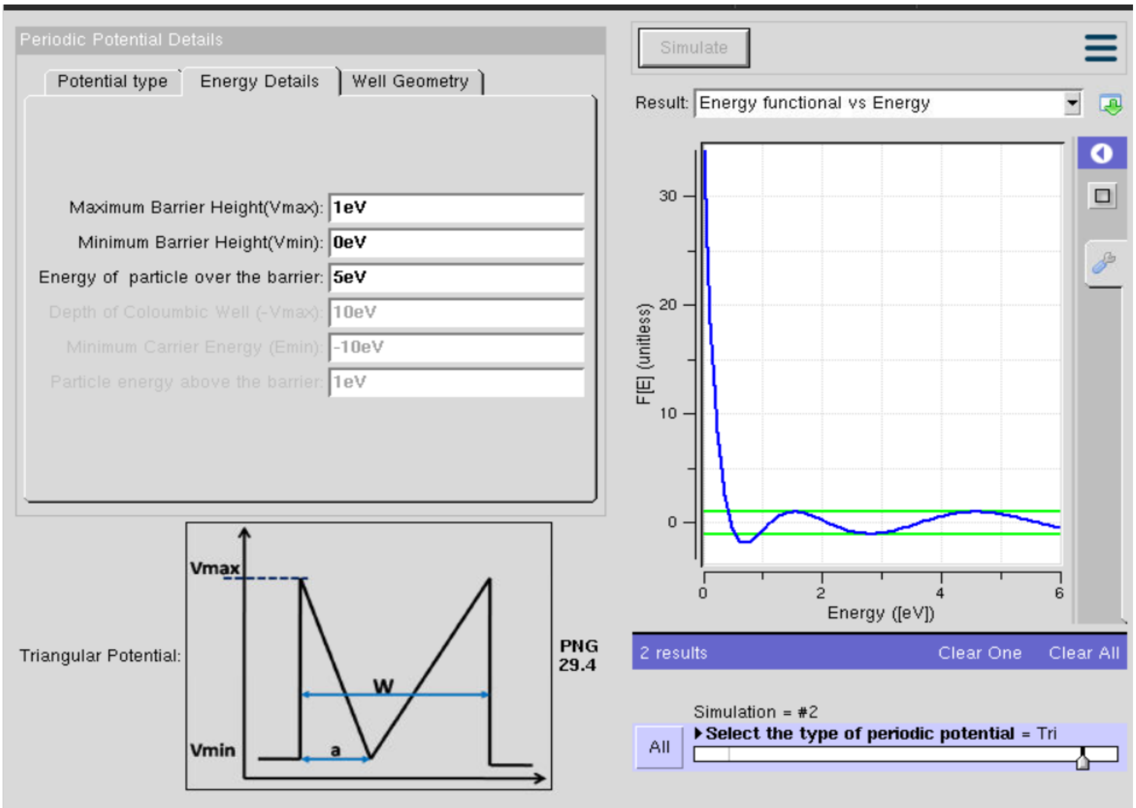


1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Result: All Eigen Energies
Eigenvalues (eV)
3
2
1
-0.1
-0.05
0
0.05
0.1
3 results
Clear One
Clear All

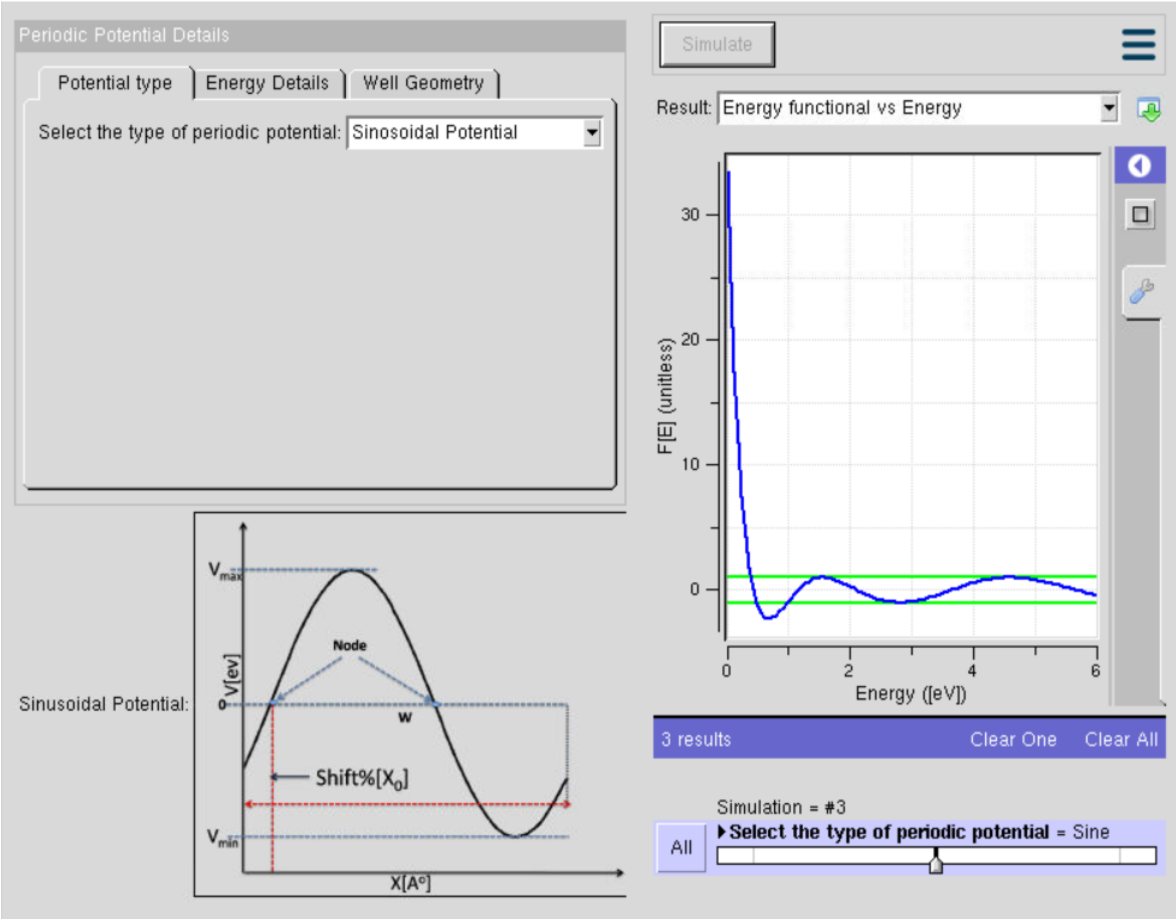
4th Experiment



Case Study 1



Case Study 2



Periodic Potential Details
Potential type
Energy Details
Well Geometry
Select the type of periodic potential:
Sinosoidal Potential
Sinusoidal Potential:
Vmax
Vmin
V[ev]
0
Node
W
Shift%[X0]
X[A°]
Simulate
Result:
Energy functional vs Energy
F[E] (unitless)
30
20
10
0
0
2
4
6
Energy ([eV])
3 results
Clear One
Clear All
Simulation = #3
All
Select the type of periodic potential = Sine